

Resumes: Answers through Wisdom and Justice

The Question

Bertrand and Mullainathan (2004) sent 4,870 fictitious resumes to help-wanted ads in Boston and Chicago newspapers, randomly assigning each resume an African-American-sounding or White-sounding name. Our data records, for each resume, whether it received a callback, along with the name's connotation and the resume's characteristics.

Imagine that you work for a civil rights organization in Chicago. You want to understand the process by which black US citizens are discriminated against in hiring today.

The civil rights organization asks: What is the average causal effect of having a black-sounding name, rather than a white-sounding name, on the probability that a job applicant in Chicago today receives a callback?

Wisdom

Preceptor Table

What are the units, outcome, and covariates for this question?

- Units:

Each row is one job application submitted in Chicago today by a black US citizen (or by anyone whose name signals a race).

- Outcomes:

Two *potential* outcomes per row — callback if the resume carries a black-sounding name, and callback if the same resume carries a white-sounding name.

- Covariates:

None are forced by the question, though quality is a natural candidate if we later ask whether the effect varies.

- Treatment (if any):

The racial connotation of the name on the resume — black-sounding versus white-sounding. Note that the treatment is the *name*, which can be manipulated, not the applicant's race, which cannot.

- Quantity of interest:

The average difference between the two potential outcomes — the average causal effect of a black-sounding name on callback probability.

Describe the key components of a Preceptor Table for this problem; Use words like “units,” “outcomes,” and “covariates.”

The Preceptor Table should have one row per job application in Chicago today. The columns are Applicant, the two potential outcomes (Callback if Black Name, Callback if White Name), and the Treatment (Name on Resume).

Preceptor Table --- Civil rights organization (causal)¹

Unit ²	Potential Outcomes ³		Treatment ⁴
Applicant	Callback if Black Name	Callback if White Name	Name on Resume
Jamal Jones	No	Yes	Black-sounding
Emily Walsh	No	Yes	White-sounding
...	...	...	...
Lakisha Washington	No	No	Black-sounding

¹ If all the information in this table were available, we could answer the question: What is the average causal effect of having a black-sounding name, rather than a white-sounding name, on the probability that a job applicant in Chicago today receives a callback?

² Each row represents one job application submitted in Chicago today. Tens of thousands of such applications are submitted each year.

³ The two potential outcomes: whether this application receives a callback if the resume carries a black-sounding name, and if it carries a white-sounding name. The cross-hatched cells mark each application's unobservable potential outcome — the one we could never see, because the resume actually carried the other kind of name.

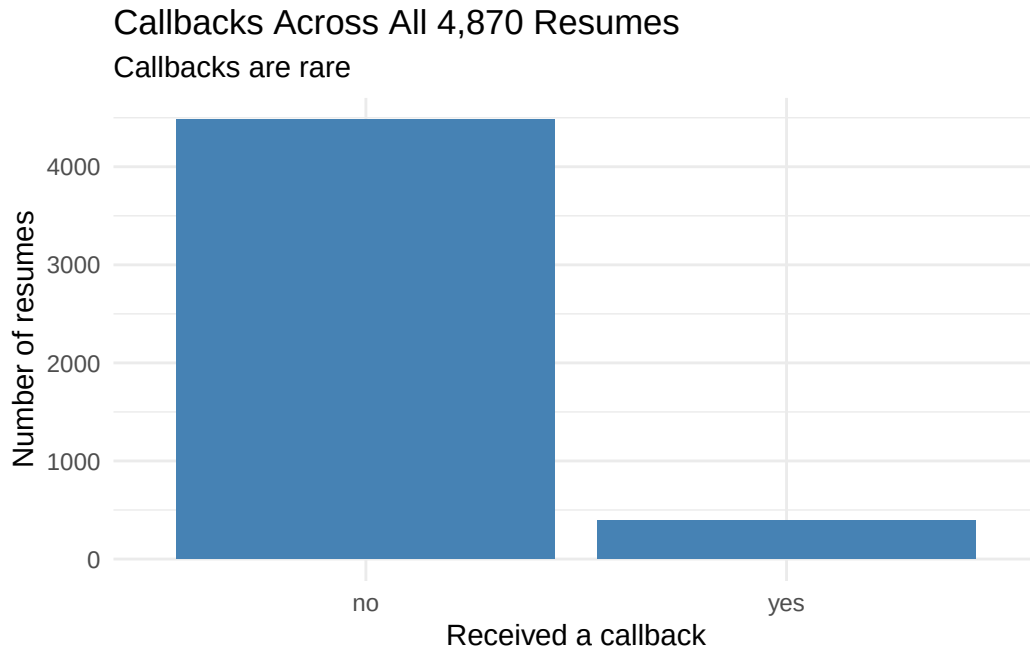
⁴ The racial connotation of the name at the top of the resume: black-sounding or white-sounding. The treatment is the name, which can be manipulated, not the applicant's race, which cannot.

Exploratory Data Analysis

Create a plot that shows the distribution of `call`.

```
resumes |>
  ggplot(aes(x = call)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Callbacks Across All 4,870 Resumes",
    subtitle = "Callbacks are rare",
    x = "Received a callback",
```

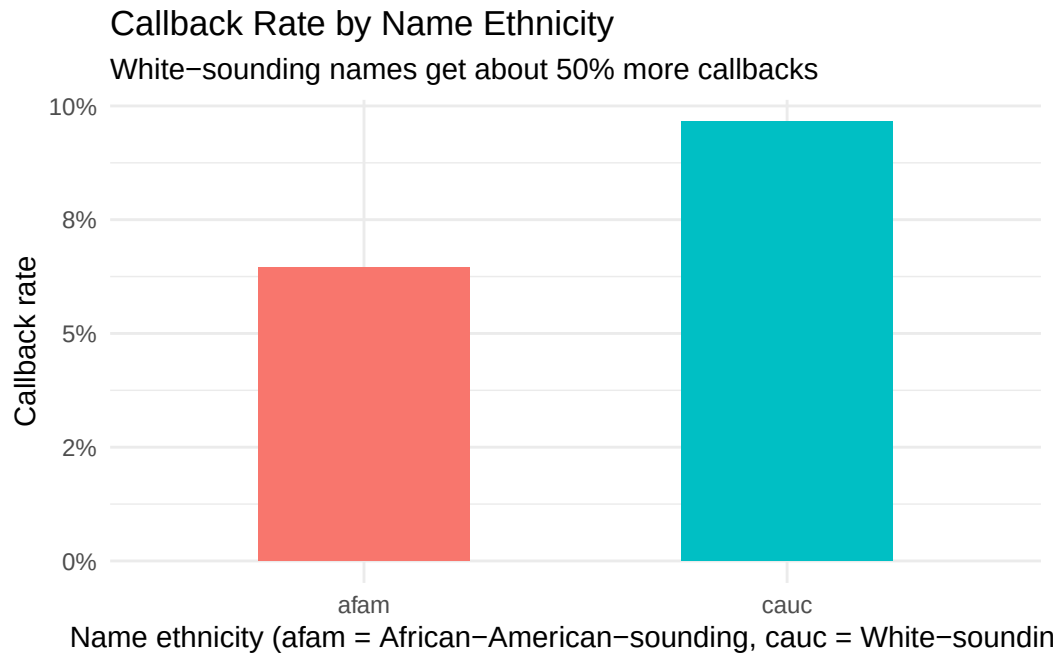
```
y = "Number of resumes"
) +
theme_minimal()
```



Callbacks are rare: only about 8% of all resumes got one. Rare binary outcomes are exactly where a Bernoulli model (Justice) is needed and where intuitions based on averages of continuous outcomes can mislead.

Create a plot that shows the relationship between `call` and `ethnicity`.

```
resumes |>
  group_by(ethnicity) |>
  summarize(callback_rate = mean(call == "yes")) |>
  ggplot(aes(x = ethnicity, y = callback_rate, fill = ethnicity)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(
    title = "Callback Rate by Name Ethnicity",
    subtitle = "White-sounding names get about 50% more callbacks",
    x = "Name ethnicity (afam = African-American-sounding, cauc = White-sounding)",
    y = "Callback rate"
  ) +
  theme_minimal()
```



Resumes with white-sounding names were called back about 9.7% of the time, against 6.5% for black-sounding names — roughly 50% more callbacks, on a perfectly balanced design of 2,435 resumes per group.

Justice

Population Table

Describe the key components of a Population Table for this problem.

One row per job application per moment in time, drawn from the population of applications submitted by people whose names signal a race in large US cities from roughly 2000 to today. The Data rows are the 2001 fictitious resumes; the Preceptor rows are Chicago applications today. The columns are Source, Applicant, Year, the two potential outcomes, and the Treatment.

Population Table --- Civil rights organization (causal) ¹					
Source	Unit/Time ²		Potential Outcomes ³		Treatment ⁴
	Applicant	Year	Callback if Black Name	Callback if White Name	Name on Resume
...	...	...	...	...	...
Data	Emily Walsh	2001	...	Yes	White-sounding
Data	Jamal Jones	2001	No	...	Black-sounding
Data	...	...	...	...	...
Data	Lakisha Washington	2001	No	...	Black-sounding
...	...	...	...	...	...
Preceptor	Jamal Jones	2026	No	Yes	Black-sounding
Preceptor	Emily Walsh	2026	No	Yes	White-sounding
Preceptor	...	...	...	...	...
Preceptor	Lakisha Washington	2026	No	No	Black-sounding
...	...	...	...	...	...

¹ This table combines the Bertrand and Mullainathan (2004) resume data with the civil rights organization's Chicago applications today, both drawn from the population of job applications carrying racially-coded names in large US cities from 2000 to today.

² Data rows are fictitious resumes sent to Boston and Chicago employers in 2001. Preceptor rows are applications submitted in Chicago in 2026. The 25-year gap is the stability concern.

³ In Data rows, only the potential outcome matching the name the resume actually carried was observed; the other is shown as '...' because no one measured it. In Preceptor rows, both potential outcomes are filled in with the truth, and the unobservable one is cross-hatched.

⁴ In Data rows, the name was randomly assigned to the resume by the experimenters. In Preceptor rows, the name is whatever the applicant actually uses. Random assignment is what makes unconfoundedness credible in the data; nothing guarantees it in the Preceptor rows.

Assumptions

Counter-example to the assumption of validity:

The data's "race" column records the race a *name* signals on a fictitious resume, but the organization ultimately cares about discrimination against actual black citizens; the connection between names and race was imperfect in 2001 and is weaker today. Likewise, a "callback" in the data is a phone message left on a voicemail box set

up for a made-up person responding to a newspaper ad; a callback for today's applicant is a response routed through an online portal, often after algorithmic resume screening. Same column names, different things.

Counter-example to the assumption of stability:

The data were collected in 2001. If discrimination in hiring has decreased (or increased, or changed form — say, moving from name screening to algorithmic screening) since then, the relationship between the name on a resume and the callback outcome today is not the relationship in our data. The civil rights organization's question is about *today*, 25 years after the data.

Counter-example to the assumption of representativeness:

The data are not a sample of any real applicant population: the resumes were invented, sent only in response to newspaper help-wanted ads, only for entry-level positions, and only in Boston and Chicago. Real applicants in Chicago today differ in occupation mix, application channel, and resume content in ways the study's design cannot reflect. Nothing about a made-up resume answering a 2001 newspaper ad guarantees it resembles the applications the organization cares about.

Counter-example to the assumption of unconfoundedness:

The experimenters randomly assigned names to resumes, so treatment assignment should be independent of the potential outcomes and unconfoundedness should hold. The honest worry is execution, not design: if the assignment process was not carried out cleanly — for example, if better resumes were disproportionately given white-sounding names — the name would be confounded with resume quality.

Probability Family and Link Function

What probability family should we use?

Bernoulli Distribution

Why should we use that probability family?

The outcome is binary — a resume either receives a callback or it does not — and the Bernoulli distribution is the distribution for a single yes/no outcome.

What is the link function for our outcome?

The logit link. Our model will be a logistic regression.